

Datasets & Benchmarks — Expanded Guide

Five resources plus an evaluation schema for studying tool-selection accuracy

1. Dataset selection principle

A dataset is valuable for this repository when it lets a researcher observe or score tool selection, retrieval, calling, planning, state dependency or multi-step agent behavior. The resources below are therefore selected for their connection to tool-use evaluation rather than for generic popularity.

1. BFCL — Berkeley Function Calling Leaderboard

Description: Function/tool-calling benchmark with single-turn, multi-turn and agentic settings.

Application to this topic: Directly measures correctness of function calls and supports structured evaluation.

Official source: <https://gorilla.cs.berkeley.edu/leaderboard>

Suggested experiment: define a candidate-tool set, run the agent, record the selected tool and compare it with the benchmark's expected tool/action or trajectory.

2. API-Bank — API-Bank

Description: Tool-augmented LLM benchmark with APIs and tool-use interactions.

Application to this topic: Useful for testing API retrieval, planning and calling.

Official source: <https://aclanthology.org/2023.emnlp-main.187/>

Suggested experiment: define a candidate-tool set, run the agent, record the selected tool and compare it with the benchmark's expected tool/action or trajectory.

3. ToolBench — ToolLLM / OpenBMB

Description: Large-scale tool-use resource built around many real-world APIs and instruction-solution trajectories.

Application to this topic: Useful for studying retrieval and selection with a large candidate-tool pool.

Official source: <https://github.com/OpenBMB/ToolBench>

Suggested experiment: define a candidate-tool set, run the agent, record the selected tool and compare it with the benchmark's expected tool/action or trajectory.

4. ToolTalk — Microsoft Research

Description: Conversational tool-use benchmark with ground-truth tool sequences.

Application to this topic: Useful for evaluating correct tool sequences across conversations.

Official source: <https://github.com/microsoft/tooltalk>

Suggested experiment: define a candidate-tool set, run the agent, record the selected tool and compare it with the benchmark's expected tool/action or trajectory.

5. ToolSandbox — Apple / ACL NAACL Findings 2025

Description: Stateful, conversational, interactive tool-use evaluation benchmark.

Application to this topic: Useful when correct tool choice depends on previous state and intermediate execution.

Official source: <https://aclanthology.org/2025.findings-naacl.65/>

Suggested experiment: define a candidate-tool set, run the agent, record the selected tool and compare it with the benchmark's expected tool/action or trajectory.

2. Example dataset schema

Field	Example meaning
task_id	Unique evaluation task
instruction	User request
candidate_tools	Tools available to the agent
expected_tool	Reference tool or tool sequence
agent_selection	Tool selected by model
arguments	Arguments generated for the tool
execution_result	Tool output
selection_correct	1/0 or graded score
notes	Failure analysis

3. Metrics to derive

- Tool Selection Accuracy
- Top-k Tool Retrieval Recall
- Argument Accuracy
- Sequence Exact Match
- Tool Call Success Rate
- State-Dependency Success
- Robustness under noise
- Abstention accuracy